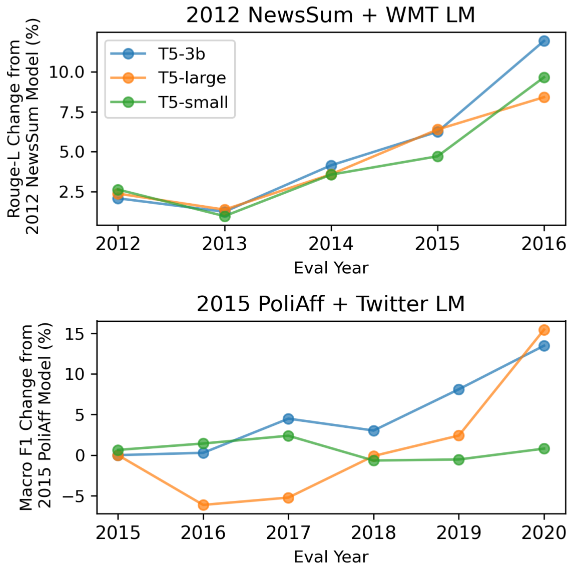

Figure 7: **Task analogies can offset downstream temporal misalignment without labeled data from the target time.** We report the performance of NewsSum and PoliAff T5 models updated using WMT LM and Twitter LM vectors for each target evaluation time. We report the percent improvement of the best updated model over 2012 NewsSum and 2015 PoliAff models on each target time for all model sizes.

small has no improvement over the baseline and T5-large task analogies perform worse than the baseline on 2016 and 2017 before improving on 2019 and 2020. Strangely, we find that only scaling α_1 can also improve performance on future years. We report these α ablations and our results on two other classification tasks in Appendix §A.6. We observe mostly similar results on these tasks, although there are task-specific inconsistencies.

4.5 Generalizing to Multiple Time Periods

Because interpolations prove useful for generalizing to intervening and future time periods, we next test if we can build models that perform well on *multiple* time periods by interpolating between all time vectors for a task.

Method We approach this problem with the *model soup* technique (Wortsman et al., 2022). One of the key practical advantages of soups is that constituent time-specific models can be trained independently (on smaller compute budgets) and combined at any time. Furthermore, the multi-year model does not need to be retrained to include new time periods; new time periods can be incorporated by merely growing the soup with additional finetuned models.

We attempt to create a multi-year model by following the recipe outlined by Wortsman et al. (2022). They introduce two soup variants: the *uniform soup* and *greedy soup*. The uniform soup applies a uniform weight among all constituent models in the interpolation, while the greedy soup is an iterative procedure that only includes models in the soup that improves validation performance. We assess both variants here.

Our "uniform time soup" is $\theta_{\text{pre}} + \frac{1}{|T|}\sum_{t\in T}\tau_t$ where T is the set of all years for a given task. For our "greedy time soup," we implement a similar algorithm to Wortsman et al. (2022) which samples time vectors (with replacement) from each year in order of decreasing performance and adds them to the average model soup if they improve performance.

To evaluate our ability to build models that generalize to multiple time periods, we measure the average performance across all evaluation years for each task. We compare our model soups against two baselines: 1) a model trained on all shuffled available data at once and 2) the best-performing model finetuned on only a single year of data. The all-year model is the most compute-intensive approach.

Results Overwhelmingly, time soups perform worse than the model finetuned on all shuffled available data. For WMT LM and NewsSum, the uniform time soup performs worse than even the best single year model, despite having access to five times the amount of finetuning data. The greedy time soup only improves over the best single-year model on PoliAff with a single macro F1 point gain. These findings suggest that a model which generalizes to multiple time periods does not lie in a region of weight space bounded by models finetuned on single years of data. Future work may explore more sophisticated methods of merging which to induce better performing multi-year models.

4.6 Summary

We propose methods for updating models to intervening, future, and multiple time periods using time vector arithmetic. We find that interpolating between two time vectors improves performance on unseen intervening times at both yearly and monthly scales. Similarly, we can improve performance on the future with unlabeled data from target times using time vector analogies. Building a multi-year model with a "soup" of time vectors,

	Perplexity (↓)	*Rouge* (↑)	*F1* (↑)
Method	WMT LM	NewsSum	PoliAff
Best single-year model	34.45	38.95	0.7101
Uniform time soup	34.70	33.05	0.6078
Greedy time soup	34.45	38.95	0.7202
Training on all years	**29.17**	**40.07**	**0.7853**

Table 3: **Interpolation does not enable generalization to multiple time periods simultaneously.** Here, we measure the average performance of models on all years. We compare multiple ways of building multi-year models; T5-small models finetuned to individual years or all years, and "time soups" created by averaging together all year time vectors for a task.

however, does not approach the performance of a model finetuned on all times at once. These results suggest that task arithmetic can be a simple way to update models to new times, but it does not help to improve genearlization across the board within a single model.

5 Related Work

Semantic Drift Although changes in the full weight spaces of models over time have not been previously explored, semantic changes in word embeddings over time are well-documented (Hamilton et al., 2016). Temporal misalignment (Bamler and Mandt, 2017; Gonen et al., 2021) and word analogies over time (Szymanski, 2017) have also been studied in embeddings. Our work extends these analyses to the full set of language model parameters.

Temporal Misalignment The phenomenon of temporal misalignment in language models has gained attention in the last three years. Moving from semantic drift to model misalignment in the twitter domain, Jaidka et al. (2018) measure gender and age classifier degradation over time, Rijhwani and Preoţiuc-Pietro (2020) demonstrate the effect of temporal drift on named entity recognition, and Loureiro et al. (2022) find decay on a variety of language modeling and downstream tasks. Lazaridou et al. (2021) extend these analyses to language modeling on News and Science domains and show that increasing model size does not help mitigate temporal misalignment. Luu et al. (2022) compare temporal misalignment across a variety of downstream tasks, finding that degradation varies greatly over both domain and task. Using the same suite of tasks, Longpre et al. (2023) report similar degradation over time in pretraining regardless of model size.

Updating LMs Recent attempts at updating language models to new time periods have used a range of techniques. Luu et al. (2022) find limited improvement with continued pretraining (Röttger and Pierrehumbert, 2021) on target times. Similar to the sequential updating setting, however, Lazaridou et al. (2021) show that dynamic evaluation (Gururangan et al., 2020) can improve language modeling performance on new times, but results in forgetting the past. More recent techniques have been proposed for keeping models up to date in the QA domain by adding flags with the year for each example (Dhingra et al., 2022) or by discarding outdated facts (Zhang and Choi, 2023). Unlike these methods, we consider the problem of updating models to new time periods without data in the target time and without additional finetuning.

Interpolation Our work draws heavily on recent techniques for editing models directly with interpolation and task analogies. Time vectors are an application of task vectors (Ilharco et al., 2023) to the time domain, our interpolation experiments are inspired by previous work on patching models for multiple tasks (Ilharco et al., 2022), and our time soups are an application of models soups (averaging multiple models trained with different initializations) (Wortsman et al., 2022).

6 Conclusion

We connect studies of temporal misalignment and weight arithmetic with time vectors, formed by finetuning a model on a specific time period and then subtracting its pretrained weights. We show that the weights of time vectors are more similar if their corresponding times are closer and vice versa. These similarities are highly correlated to temporal misalignment at both yearly and monthly scales (which exhibit seasonal patterns). Leveraging this temporal structure in weight space, we induce new models that perform better on intervening years by interpolating between adjacent time vectors. Similarly, we use task analogies to improve downstream performance on future time periods using only unlabeled data from those times. These results show that task arithmetic can be a simple tool for combating temporal misalignment.

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